Common Functionalities

1. Registration sign in, Sign up
2. Multiple user login (customer, admin[shop owners/business owners], superadmin etc) and different landing page/Home page
3. Filtering options such as: Using drop down menu customer can find out their required services/products
    a. such as: car rental - filter based on price, seat number, location and availability
    b. Such as: filter using dates, review rating, car model etc
4. Customer can give reviews/ratings/comments
5. Super admin can delete any shop owners if needed based on customer reviews
6. Cart (add item, delete item, purchase item), payment
7. admins (Shop owners/business owners) can see their earnings/sell (who bought their product, date and time, price, total earnings etc)
8. Stock or inventory dashboard (shop owners can see how many products are sold and how many are left in the stock), new item can be added
9. Customer can see different special packages or discount offers

How the system will operate

Customers (n Number of customers) **1st party**	⇒ ⇒ ⇒	**IT system provides 3rd party (we are CS Engineers) connecting customers with business owners**	⇐ ⇐ ⇐	Business owners (n Number of business owners/service providers) **2nd party**

1. We will assume system like this connecting customers with business/service providers such as:
    a. pathau, Uber
    b. foodpanda, panda mart, chaldal
    c. bikroy, daraz

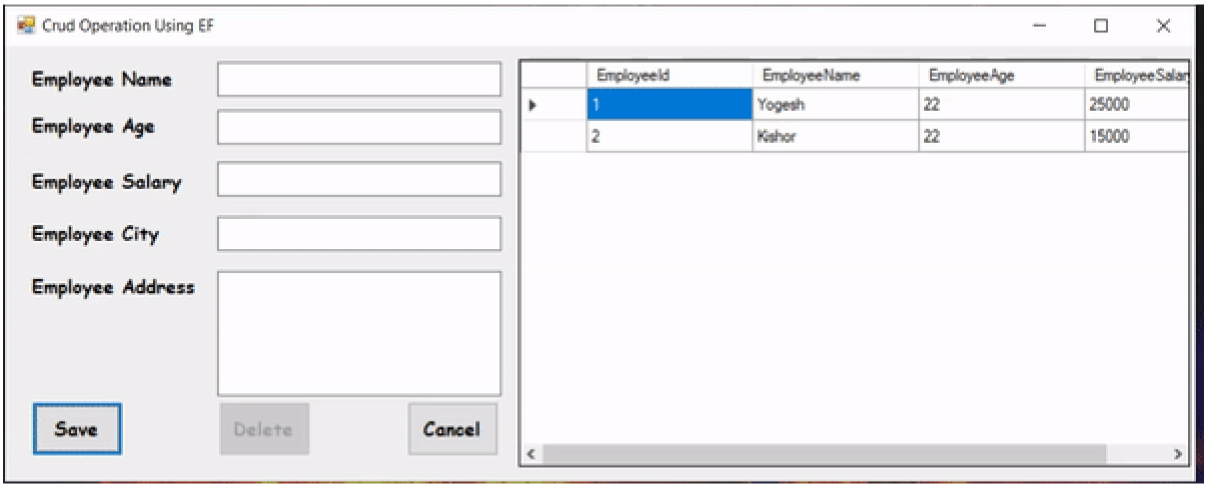
2. **As a IT support (web/app/software) provider we will get commission from the business owners**

Multiple user Login **(User Table)** - querying user_type we will know what type of user and we will land them into separate pages/forms			
User_type	ID	Password	details
admin/superadmin/customer			

To Implement we have to do learn **CRUD Datagridview very well**

https://www.techieclues.com/articles/insert-update-and-delete-records-in-datagridview-c-sharp
https://www.c-sharpcorner.com/article/crud-operation-windows-form-app-using-entity-framework/

Using the following interface we can able to develop all the functionalities required for our application:



We want to get your report first –

Tasks

1. Prepare a report like https://github.com/Korbanul/Travel_World-.NET
2. Must need to upload your report into github in README.md file
    a. **How to prepare a README.md file for github**,
        i. prepare a report and ask chatgpt to convert it to README.md format for github. ChatGPT knows very well how to do that in seconds.
3. You have to **submit individually CRUD task** to me but **report writing and project submission is a group task**
4. You have to **prepare a short screen record video presentation** for your Tasks. To do a screen recording video my favorite is chrome web apps store https://chromewebstore.google.com/ . You can use any screen recorder tool you like. **Voice is mandatory for video recording and demonstration**.
5. Share me video recording link uploaded in youtube/google drive/one drive etc.
6. Within 1 week you have to submit individual tasks (send github link and video link)
7. Two weeks time report submission (in github README.md file)
8. final project display after the final term examination within 2 days after the exam (send github link and video link of your whole project demonstration)

AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2024-2025

Section: [G]

Group No: 07

Project Report On

*Project Name [**Music Player]]***

Supervised By

Kazi Sadia

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Saiful Islam	**23-51111-1**
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Submitted By:

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				**Total =**
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

Chapter 1: Introduction

This project implements a PC Component Store Resource Management and Billing System. In this system, there are three types of users- Admin, Resource Manager, and Salesman. An Admin can perform CRUD operations on users. In the same way, a Resource Manager can perform CRUD operations on Items. A Salesman can view existing items and add specific items to a cart. When finalized, the items included in the cart are totaled and a bill is stored within the database system, as well as the total item count is updated after a transaction. Similarly, all new items and user additions and removals are reflected on the Database.

Chapter 2: User stories

- ### Admin:

I. **Login:** Admins can log in to the system by providing their user ID and password. When the credentials match, the user will be able to access the system.

II. **View and Edit Profile:** Admins can view their own profile. And can edit only his name, phone no, email, and address.

III. **User add and removal:** Admins can add new users as Admin, Manager, and Salesmen. Admins can also update their all information as well as remove them from the server.

IV. **Update password:** The admins can update their password from a specific form when the “Update Password” button is pressed. After entering the current password and providing a matching set of a new password and a confirmed new password, the password for the user will be updated.

V. **View items:** Admins can view the currently available items list. The list will include the item ID, item type, item’s brand, item’s model no, item’s price, and the quantity ready to be sold for each item.

VI. **View Transections:** Admins can view the transactions that have been done by the salesman in a window when admins click on the view transaction button. There they can see Transaction ID, Customer name, Bill ID, Total amount, Total price, and Salesman ID.

- ### Salesman:

I. **Login:** A salesman can log in to the system by providing their credentials into the system.

II. **Update password:** The user can update their password from a specific window when the “Update Password” button is pressed. After entering the current password and providing a matching set of a new password and a confirmed new password, the password for the user will be updated.

III. **View items:** A salesman can view the currently available items list. The list will include the item id, manufacturer, model, individual price and the quantity ready to be sold for each item.

IV. **Add items to cart:** A salesman can add items to the cart, by clicking on the corresponding item id from the total items list. The amounts are synced in total item view as new items are added to the cart.

V. **Update Cart:** A salesman can update the cart, which consists of a list of items a customer wants to buy. By clicking on the desired item id from the items list, a new entry is created on the cart, where the amount can be updated as per the demand of the customer.

VI. **Delete items from cart:** As needed by the customer, the salesman can remove entries entirely from the cart. When doing so, the available item count for each entry are restored in the total items list.

VII. **Confirming purchase:** After a customer finalized the cart items, the salesman needs to insert the customer's name & contact into the system. After than a receipt window is opened, where each individual item and the total bill is shown. The item list for each bill is stored within a text file in the device's memory.

VIII. **Log out:** The user can log out of the system by pressing the designated button, so that another user can log in.

- ## Manager:

I. **Log in:** Manager can log into the system by providing their credentials into the System.

II. **View Items:** Manager has accessibility to view the items in a table. The table provides a view of the items with Item Id, type, brand, model, price, stock status and the minimum stock.

III. **Add Items:** Manager can add new items in the inventory. By providing all the necessary information and pressing "add" the information shall be stored in the system.

IV. **Update Items:** Manager can update the item inventory by double clicking on the id. The existing data of the clicked item will be shown upon this operation. The manager can update the texts on necessary fields which data they want to update. Upon fulfilling all the necessary details, by pressing the ''add'' button, data will be updated.

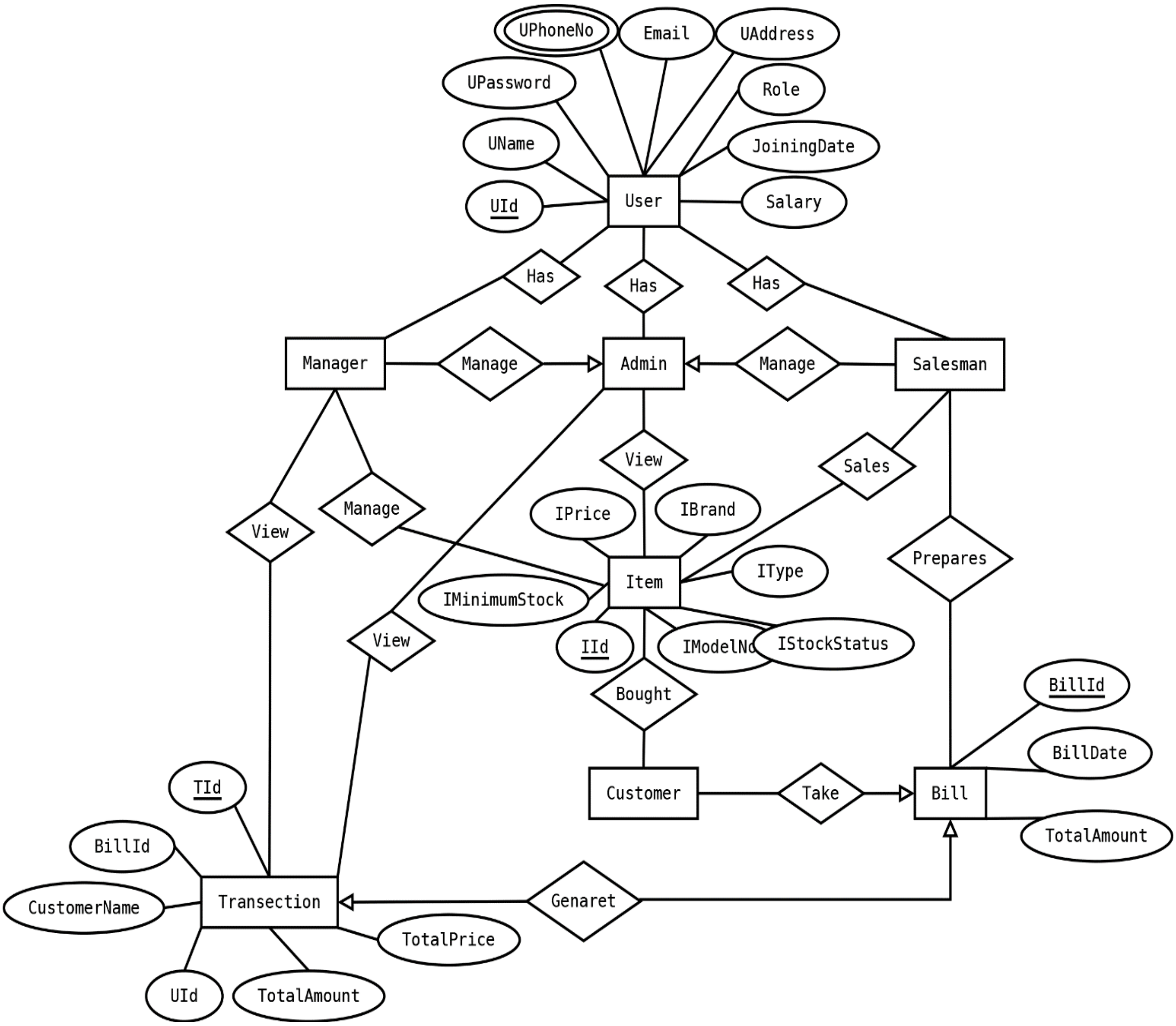
V. **Delete Items:** Manager can also perform delete operation by just only double clicking the row which they want to delete and then clicking on ''delete'' button**.**

VI. **Transaction View:** Manager can have view of transaction table by searching transaction ID, however Manager isn't authorized to make any changes on the table.

VII. **Change Profile:** Manager also has the authority to Update their Information, they could change their name, phone number, address, email, as well as password. After full filling all their information by pressing the update button information would be recorded.

Chapter 3: Database**:**

ER- Diagram:



1. **User: __UId__**, UName, UPassword, UPhoneNo, Email, UAddress, Role, JoiningDate, Salary
2. **Item: __IId__**, IType, IBrand, IModelNo, IPrice, IStockStatus, IMinimumStock
3. **Transection: __TId__**, **BillId**, **UId**, TotalAmount
4. **Bill: __BillId__**, BillDate, TotalAmount

1. User

Column	Data Type	Constraint
UId	INT	PK, Auto Increment
UName	VARCHAR(100)	NOT NULL
UPassword	VARCHAR(100)	NOT NULL
UPhoneNo	VARCHAR(20)	
Email	VARCHAR(100)	UNIQUE
UAddress	VARCHAR(200)	
Role	VARCHAR(50)	NOT NULL
JoiningDate	DATE	
Salary	DECIMAL(10,2)	

2. Item

Column	Data Type	Constraint
IId	INT	PK, Auto Increment
IType	VARCHAR(50)	NOT NULL
IBrand	VARCHAR(50)	
IModelNo	VARCHAR(50)	
IPrice	DECIMAL(10,2)	
IStockStatus	VARCHAR(50)	
IMinimumStock	INT	

3. Bill

Column	Data Type	Constraint
BillId	INT	PK, Auto Increment
BillDate	DATE	NOT NULL
TotalAmount	DECIMAL(10,2)	

4. Transaction

Column	Data Type	Constraint
TId	INT	PK, Auto Increment
BillId	INT	FK → Bill(BillId)
UId	INT	FK → User(UId)
TotalAmount	DECIMAL(10,2)	

✅ **Notes / Relationships:**

1. A **User** can have different roles: Manager, Admin, Salesman. You can manage roles via Role column.
2. **Admin/Manager** can manage/view Items.
3. **Salesman** prepares Bill for Customers.
4. Each **Transaction** links a User (the one processing the transaction) with a Bill.
5. Each **Bill** can include multiple Items (you may need a **BillItem** junction table if you want to track item-level purchases).

SQL script Based on functionalities:

System Functionalities with SQL Queries

1. User Registration & Authentication

Sign Up (Registration)

```
INSERT INTO User1 (UId, UName, UPassword, UPhoneNo, Email, UAddress, Role, JoiningDate, Salary)
VALUES ('C-001', 'Customer1', 'pass123', '01711223344', 'customer1@gmail.com', 'Banani, Dhaka', 'Customer',
GETDATE(), 0);
```

Sign In (Login)

```
SELECT UId, UName, Role
FROM User1
WHERE Email = 'customer1@gmail.com' AND UPassword = 'pass123';
```

Update Password

```
UPDATE User1
SET UPassword = 'newPass456'
WHERE UId = 'C-001';
```

2. Role-Based User Access

Check Role After Login (Landing Page Decision)

```
SELECT Role
FROM User1
WHERE UId = 'C-001';
```

- If Role = Customer → Customer home page
- If Role = Manager/Admin → Shop owner dashboard
- If Role = Superadmin → Superadmin dashboard

3. Advanced Product/Service Filtering

Filter Items by Price Range

```
SELECT *
FROM Item
WHERE IPrice BETWEEN 5000 AND 25000;
```

Filter Items by Availability (Stock > 0)

```
SELECT *
FROM Item
WHERE IStockStatus > 0;
```

Filter by Brand & Type

```
SELECT *
FROM Item
WHERE IType = 'Processor' AND IBrand = 'AMD';
```

Sort by Price Low → High

```
SELECT * FROM Item ORDER BY IPrice ASC;
```

4. Customer Feedback System

(Requires new table: Review)

```
CREATE TABLE Review(
  ReviewId VARCHAR(30) PRIMARY KEY,
  UId VARCHAR(30) FOREIGN KEY REFERENCES User1(UId),
  IId VARCHAR(30) FOREIGN KEY REFERENCES Item(IId),
  Rating INT CHECK (Rating BETWEEN 1 AND 5),
  Comment VARCHAR(200),
  ReviewDate DATE
);
```

Add Review

```
INSERT INTO Review (ReviewId, UId, IId, Rating, Comment, ReviewDate)
VALUES ('R-001', 'C-001', 'i-001', 5, 'Great processor!', GETDATE());
```

View Reviews for an Item

```
SELECT U.UName, R.Rating, R.Comment, R.ReviewDate
FROM Review R
JOIN User1 U ON R.UId = U.UId
WHERE IId = 'i-001';
```

5. Superadmin Control Panel
Delete Shop Owner (Admin)

```
DELETE FROM User1
WHERE UId = 'M-001' AND Role = 'Manager';
```

Find Low-Rated Shop Owners

```
SELECT U.UId, U.UName, AVG(R.Rating) AS AvgRating
FROM Review R
JOIN Item I ON R.IId = I.IId
JOIN User1 U ON U.UId = 'M-001'
GROUP BY U.UId, U.UName
HAVING AVG(R.Rating) < 2.5;
```

---

6. Shopping Cart & Payment

(Requires new table: Cart)

```
CREATE TABLE Cart (
   CartId VARCHAR(30) PRIMARY KEY,
   UId VARCHAR(30) FOREIGN KEY REFERENCES User1(UId),
   IId VARCHAR(30) FOREIGN KEY REFERENCES Item(IId),
   Quantity INT,
   AddedDate DATE
);
```

Add Item to Cart

```
INSERT INTO Cart (CartId, UId, IId, Quantity, AddedDate)
VALUES ('CT-001', 'C-001', 'i-001', 2, GETDATE());
```

Delete Item from Cart

```
DELETE FROM Cart WHERE CartId = 'CT-001';
```

Purchase (Generate Bill & Transaction)

```
-- Step 1: Create Bill
INSERT INTO Bill (BillId, BillDate, TotalAmount)
VALUES ('b-006', GETDATE(), 30000);

-- Step 2: Insert Transaction
INSERT INTO Transection1 (TId, CustomerName, BillId, TotalAmount, TotalPrice, UId)
VALUES ('t-006', 'Customer1', 'b-006', 2, 30000, 'C-001');
```

---

7. Sales & Earnings Dashboard for Shop Owners

View Earnings by Shop Owner

```
SELECT U.UName AS ShopOwner, SUM(T.TotalPrice) AS TotalEarnings
FROM Transection1 T
JOIN User1 U ON T.UId = U.UId
WHERE U.Role = 'Salesman'
GROUP BY U.UName;
```

Detailed Sales Report

```
SELECT T.TId, T.CustomerName, T.BillId, T.TotalAmount, T.TotalPrice, T.UId, B.BillDate
FROM Transection1 T
JOIN Bill B ON T.BillId = B.BillId
WHERE T.UId = 'S-001';
```

---

8. Stock & Inventory Management

View Current Stock

```
SELECT IId, IType, IBrand, IModelNo, IStockStatus
FROM Item;
```

Low Stock Alert

```
SELECT IId, IType, IBrand, IStockStatus, IMinimumStock
FROM Item
WHERE IStockStatus < IMinimumStock;
```

Add New Item

```
INSERT INTO Item (IId, IType, IBrand, IModelNo, IPrice, IStockStatus, IMinimumStock)
VALUES ('i-006', 'GPU', 'NVIDIA', 'RTX 4060', 55000, 6, 2);
```

---

9. Special Packages & Discount Offers

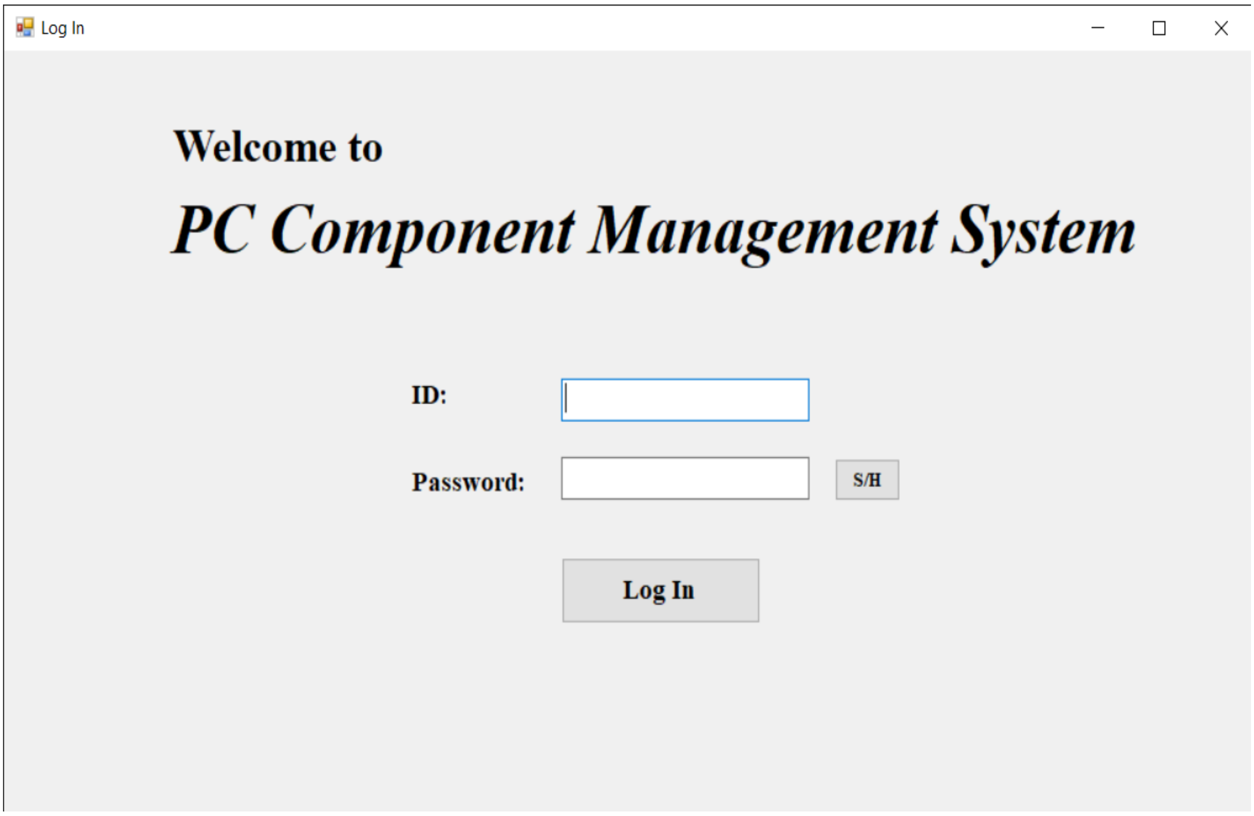
(Requires new table: Offer)

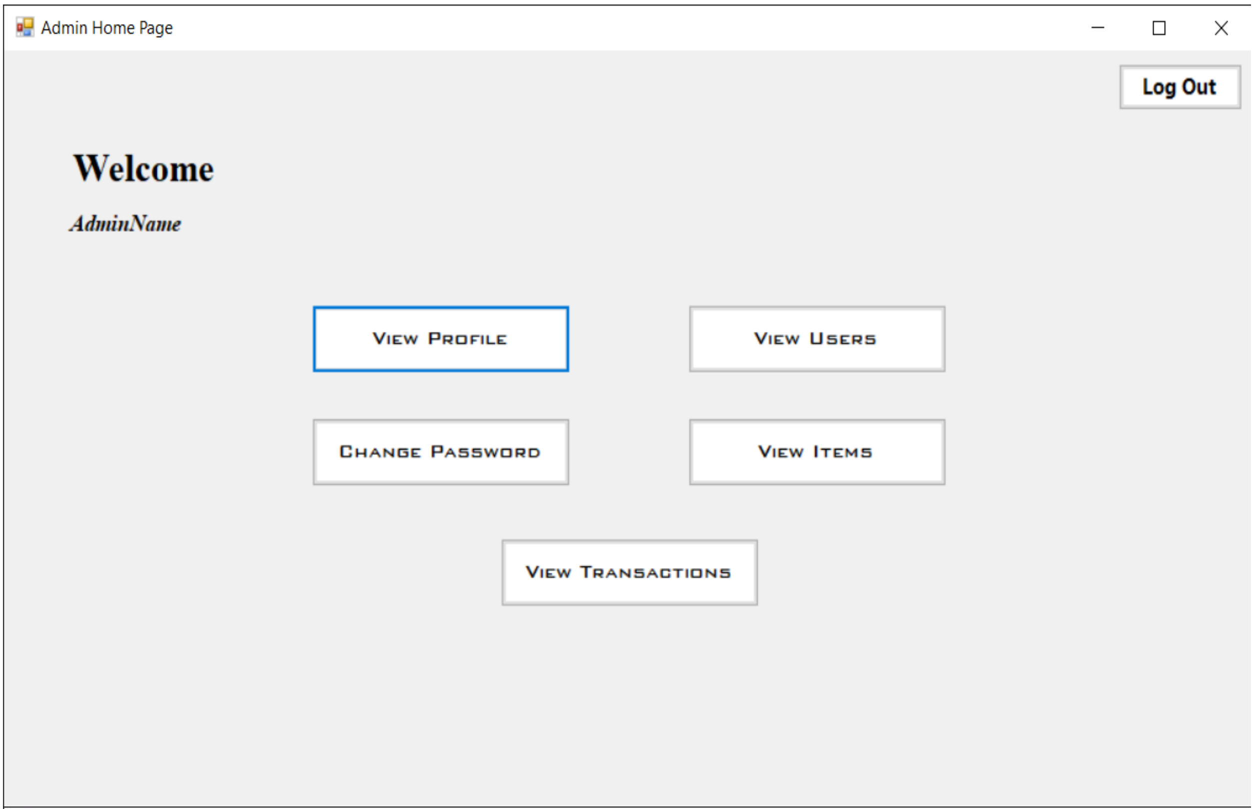
```
CREATE TABLE Offer(
```

```
  OfferId VARCHAR(30) PRIMARY KEY,
  IId VARCHAR(30) FOREIGN KEY REFERENCES Item(IId),
  DiscountPercent FLOAT,
  StartDate DATE,
  EndDate DATE
);
```
Add Offer
```
INSERT INTO Offer (OfferId, IId, DiscountPercent, StartDate, EndDate)
VALUES ('O-001', 'i-001', 10, '2023-12-24', '2023-12-31');
```
View Available Offers
```
SELECT O.OfferId, I.IType, I.IBrand, I.IModelNo, I.IPrice,
      (I.IPrice - (I.IPrice * O.DiscountPercent / 100)) AS DiscountedPrice
FROM Offer O
JOIN Item I ON O.IId = I.IId
WHERE GETDATE() BETWEEN O.StartDate AND O.EndDate;
```

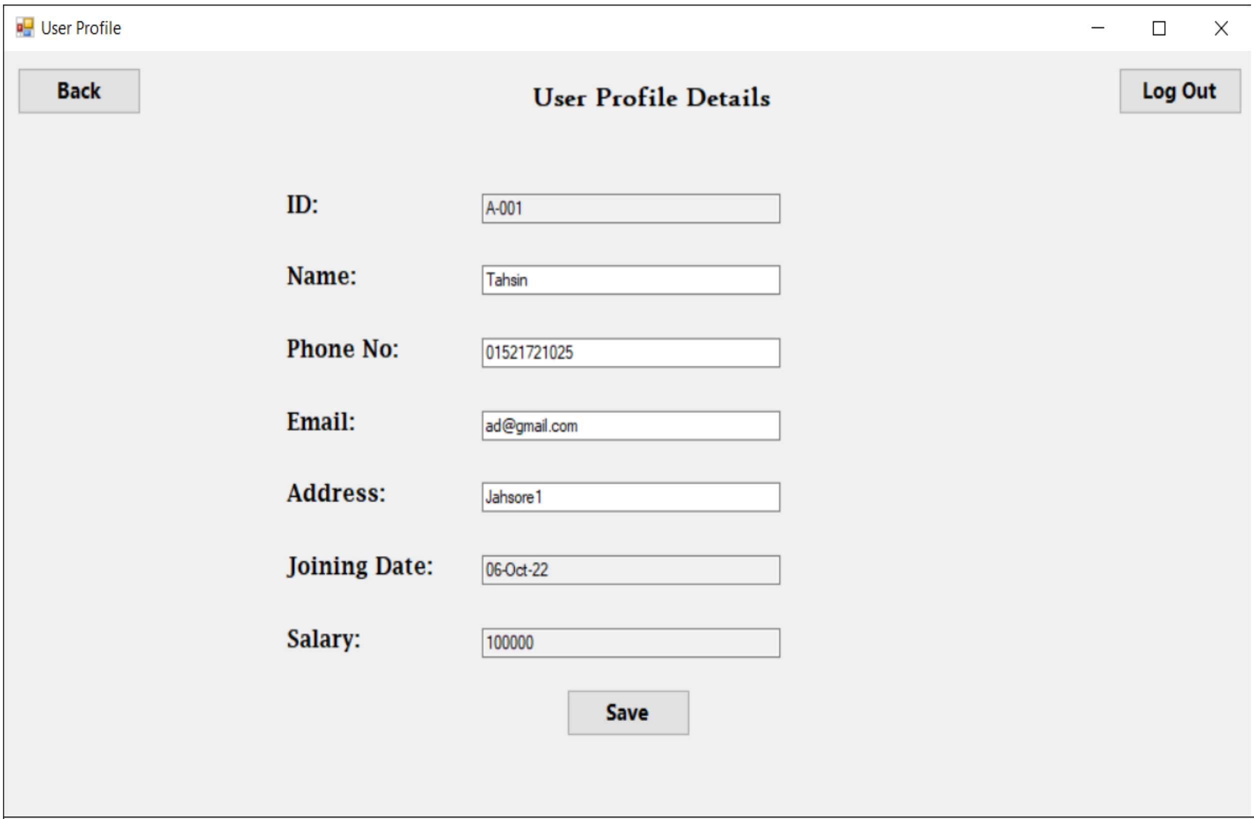
Chapter 4: Screenshot of forms**:**

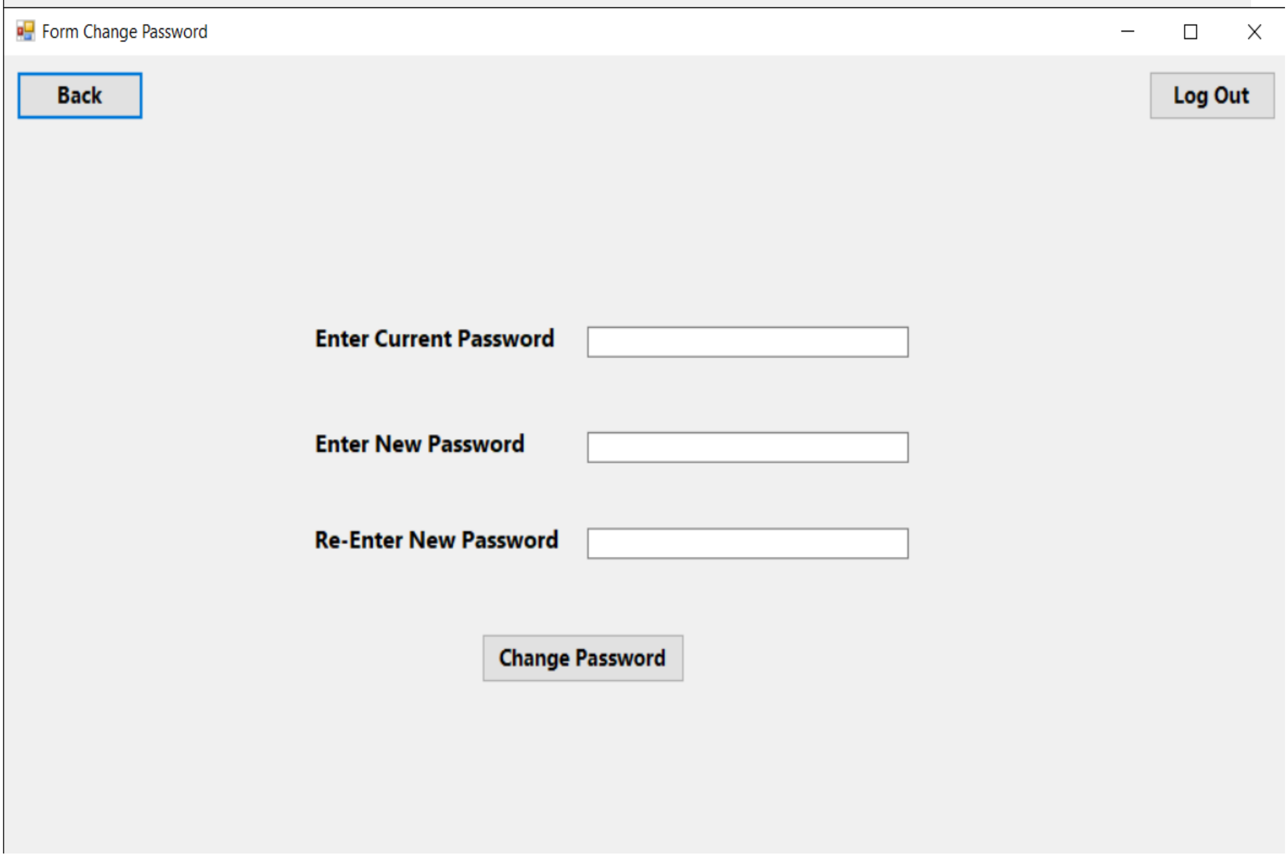
- **22-46966-1:**

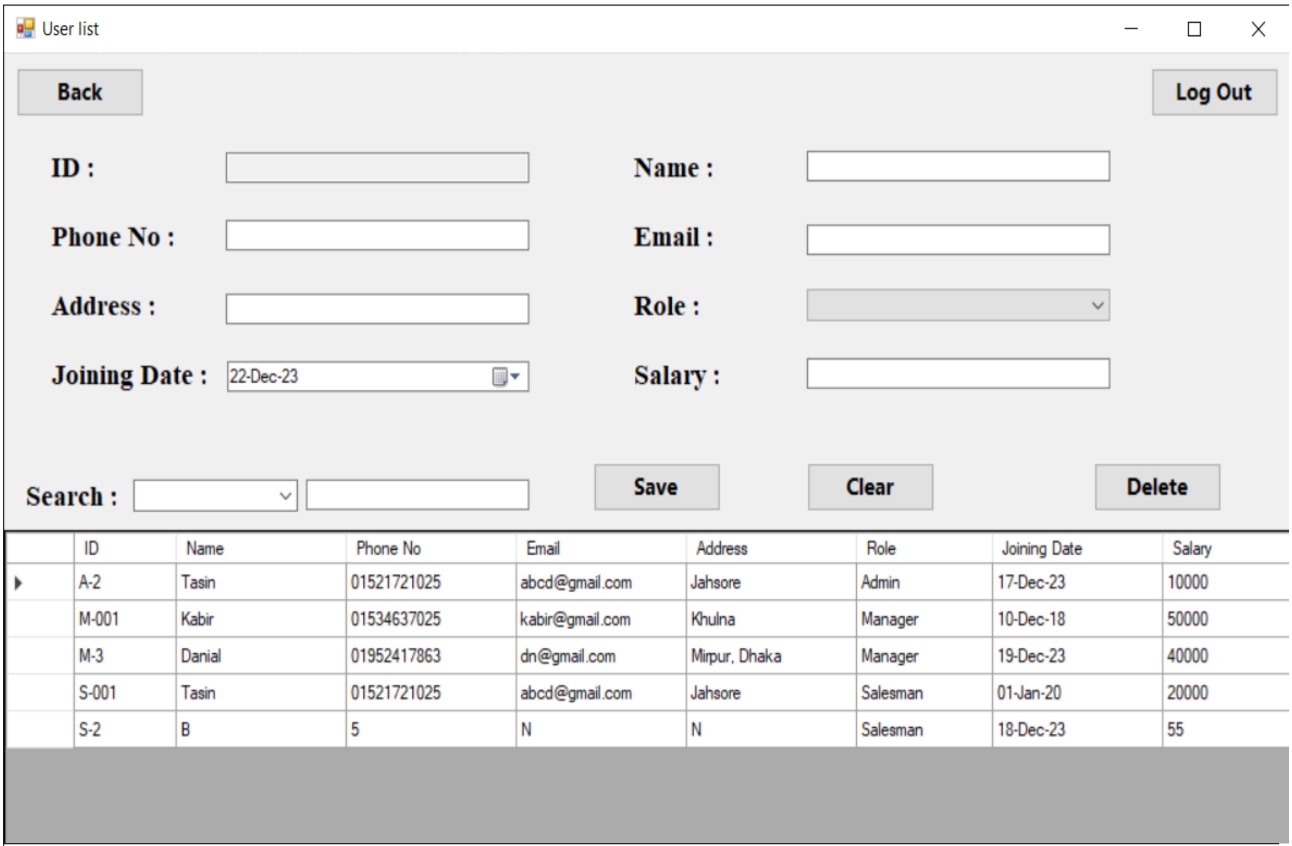



Admin Home Page
Log Out
Welcome
AdminName
View Profile
View Users
Change Password
View Items
View Transactions

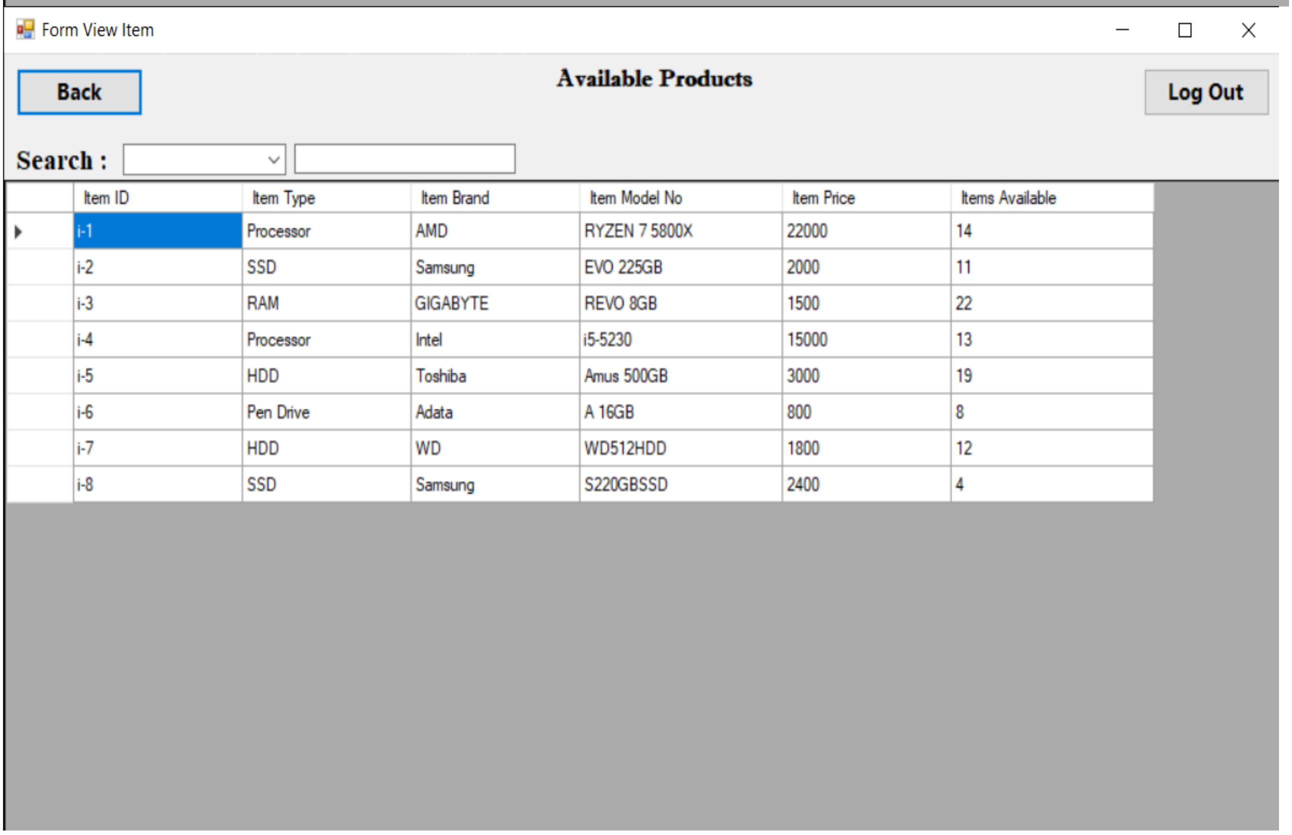

User Profile
Back
User Profile Details
Log Out
ID:
A-001
Name:
Tahsin
Phone No:
01521721025
Email:
ad@gmail.com
Address:
Jahsore1
Joining Date:
06-Oct-22
Salary:
100000
Edit


User Profile
Back
User Profile Details
Log Out
ID:
A-001
Name:
Tahsin
Phone No:
01521721025
Email:
ad@gmail.com
Address:
Jahsore1
Joining Date:
06-Oct-22
Salary:
100000
Save

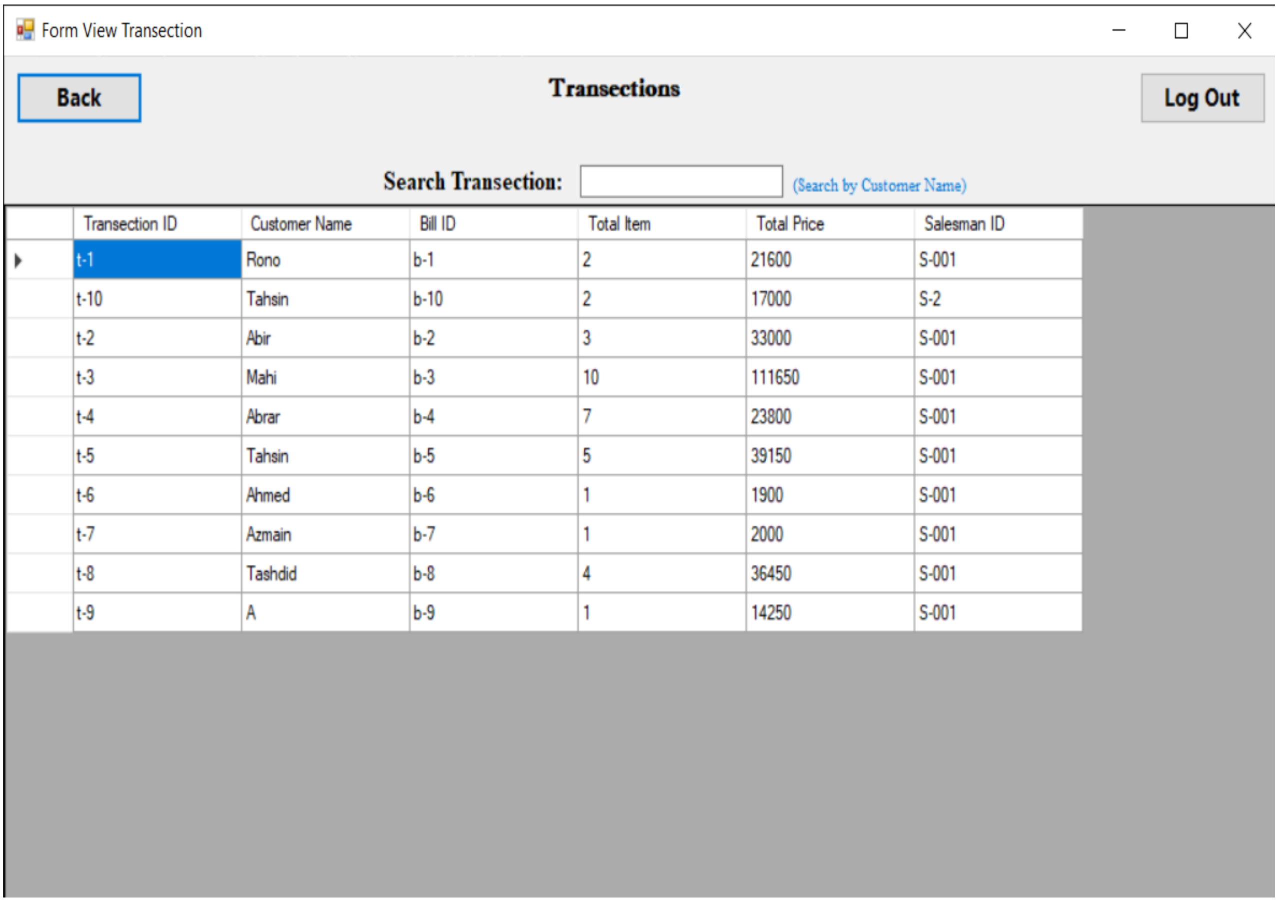

Form Change Password
Back
Log Out
Enter Current Password
Enter New Password
Re-Enter New Password
Change Password


User list
Back
Log Out
ID :
Name :
Phone No :
Email :
Address :
Role :
Joining Date :
22-Dec-23
Salary :
Search :
Save
Clear
Delete

ID	Name	Phone No	Email	Address	Role	Joining Date	Salary
A-2	Tasin	01521721025	abcd@gmail.com	Jahsore	Admin	17-Dec-23	10000
M-001	Kabir	01534637025	kabir@gmail.com	Khulna	Manager	10-Dec-18	50000
M-3	Danial	01952417863	dn@gmail.com	Mirpur, Dhaka	Manager	19-Dec-23	40000
S-001	Tasin	01521721025	abcd@gmail.com	Jahsore	Salesman	01-Jan-20	20000
S-2	B	5	N	N	Salesman	18-Dec-23	55

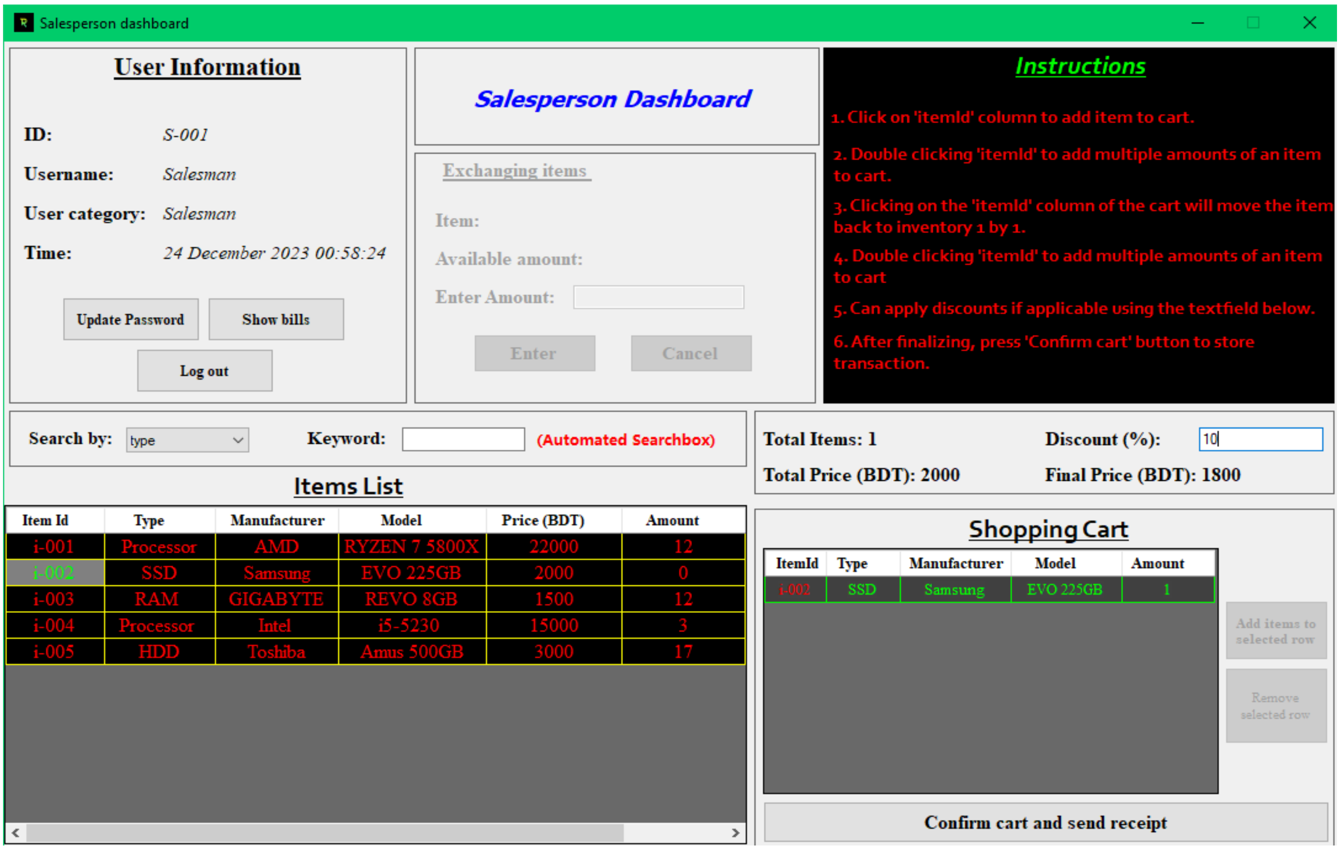

Form View Item
Back
Available Products
Log Out
Search :

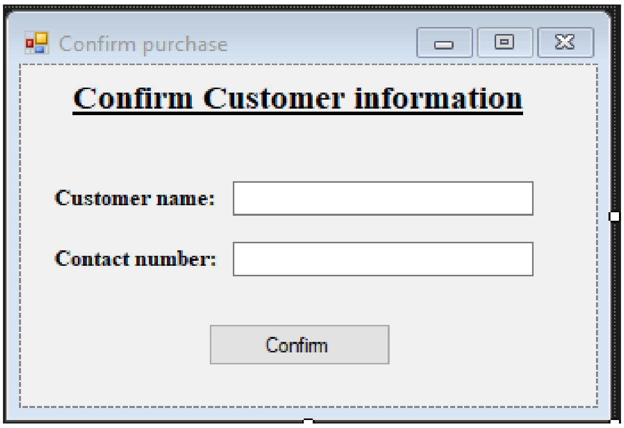
Item ID	Item Type	Item Brand	Item Model No	Item Price	Items Available
i-1	Processor	AMD	RYZEN 7 5800X	22000	14
i-2	SSD	Samsung	EVO 225GB	2000	11
i-3	RAM	GIGABYTE	REVO 8GB	1500	22
i-4	Processor	Intel	i5-5230	15000	13
i-5	HDD	Toshiba	Amus 500GB	3000	19
i-6	Pen Drive	Adata	A 16GB	800	8
i-7	HDD	WD	WD512HDD	1800	12
i-8	SSD	Samsung	S220GBSSD	2400	4

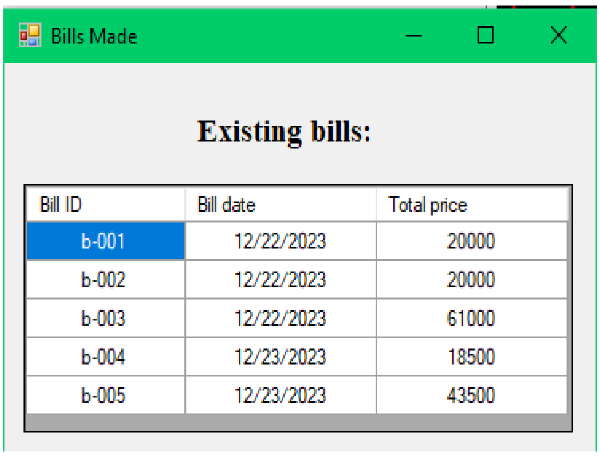


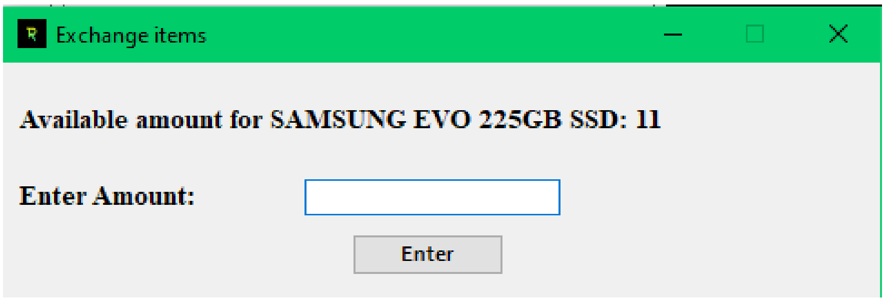
Transection ID	Customer Name	Bill ID	Total Item	Total Price	Salesman ID
t-1	Rono	b-1	2	21600	S-001
t-10	Tahsin	b-10	2	17000	S-2
t-2	Abir	b-2	3	33000	S-001
t-3	Mahi	b-3	10	111650	S-001
t-4	Abrar	b-4	7	23800	S-001
t-5	Tahsin	b-5	5	39150	S-001
t-6	Ahmed	b-6	1	1900	S-001
t-7	Azmain	b-7	1	2000	S-001
t-8	Tashdid	b-8	4	36450	S-001
t-9	A	b-9	1	14250	S-001

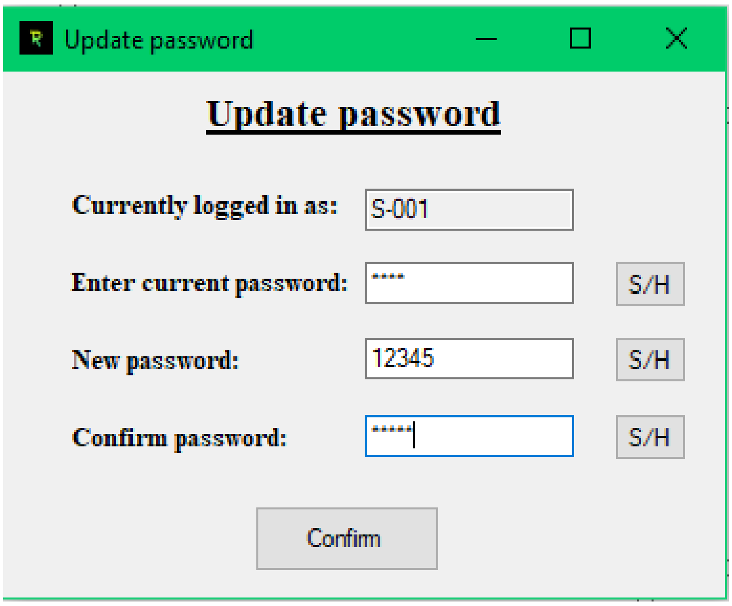
- **22-47226-1:**

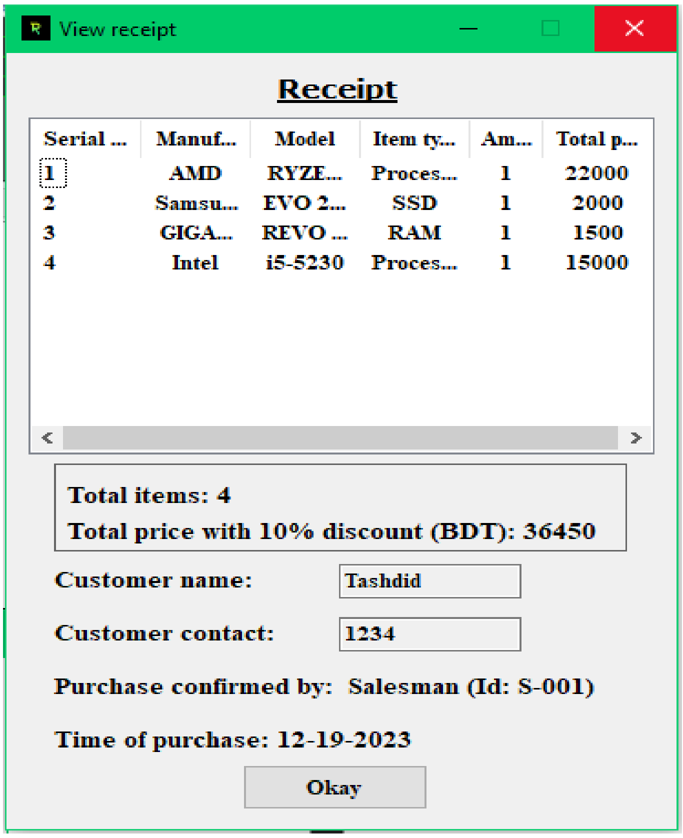

Salesperson dashboard
User Information
ID: S-001
Username: Salesman
User category: Salesman
Time: 24 December 2023 00:58:24
Update Password
Show bills
Log out
Salesperson Dashboard
Exchanging items
Item:
Available amount:
Enter Amount:
Enter
Cancel
Instructions
1. Click on 'itemId' column to add item to cart.
2. Double clicking 'itemId' to add multiple amounts of an item to cart.
3. Clicking on the 'itemId' column of the cart will move the item back to inventory 1 by 1.
4. Double clicking 'itemId' to add multiple amounts of an item to cart
5. Can apply discounts if applicable using the textfield below.
6. After finalizing, press 'Confirm cart' button to store transaction.
Search by: type
Keyword:
(Automated Searchbox)
Items List
Item Id Type Manufacturer Model Price (BDT) Amount
i-001 Processor AMD RYZEN 7 5800X 22000 12
i-002 SSD Samsung EVO 225GB 2000 0
i-003 RAM GIGABYTE REVO 8GB 1500 12
i-004 Processor Intel i5-5230 15000 3
i-005 HDD Toshiba Amus 500GB 3000 17
Total Items: 1
Discount (%): 10
Total Price (BDT): 2000
Final Price (BDT): 1800
Shopping Cart
ItemId Type Manufacturer Model Amount
i-002 SSD Samsung EVO 225GB 1
Add items to selected row
Remove selected row
Confirm cart and send receipt


Confirm purchase
Confirm Customer information
Customer name:
Contact number:
Confirm

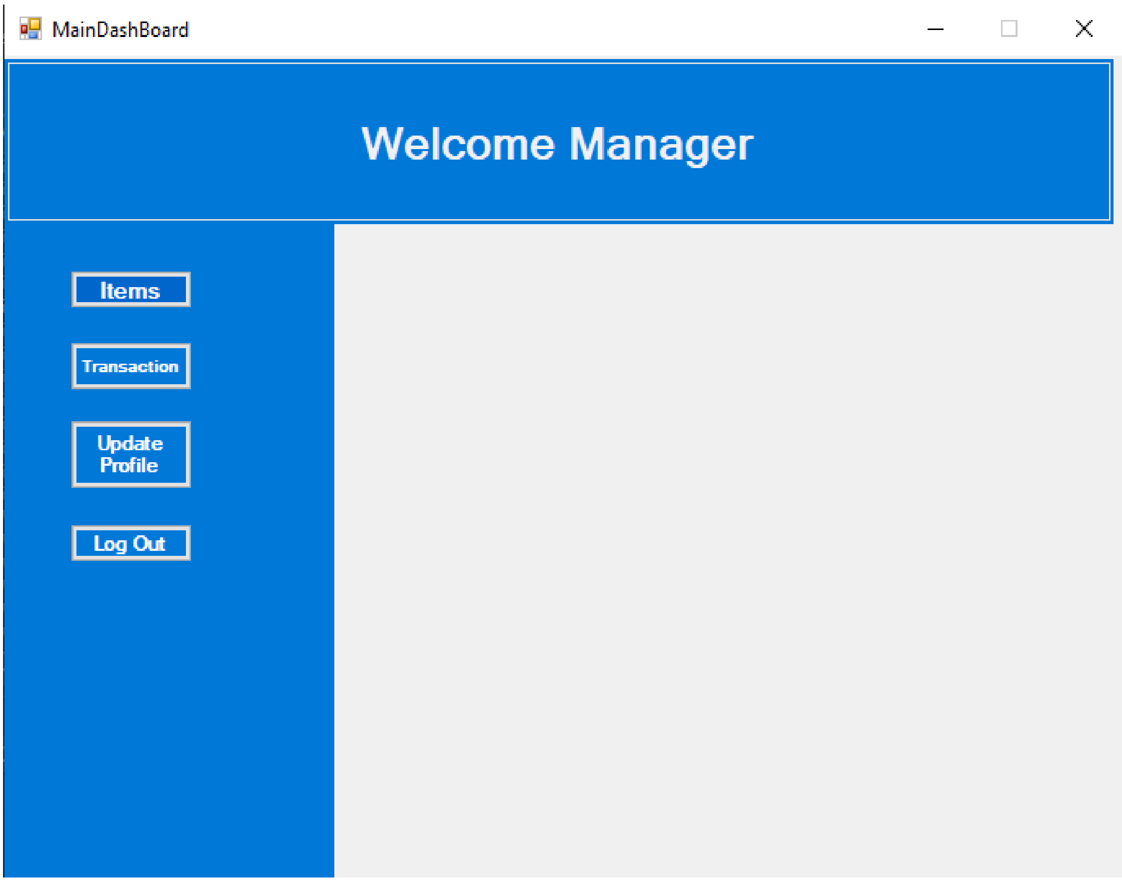

Bills Made
Existing bills:
Bill ID Bill date Total price
b-001 12/22/2023 20000
b-002 12/22/2023 20000
b-003 12/22/2023 61000
b-004 12/23/2023 18500
b-005 12/23/2023 43500

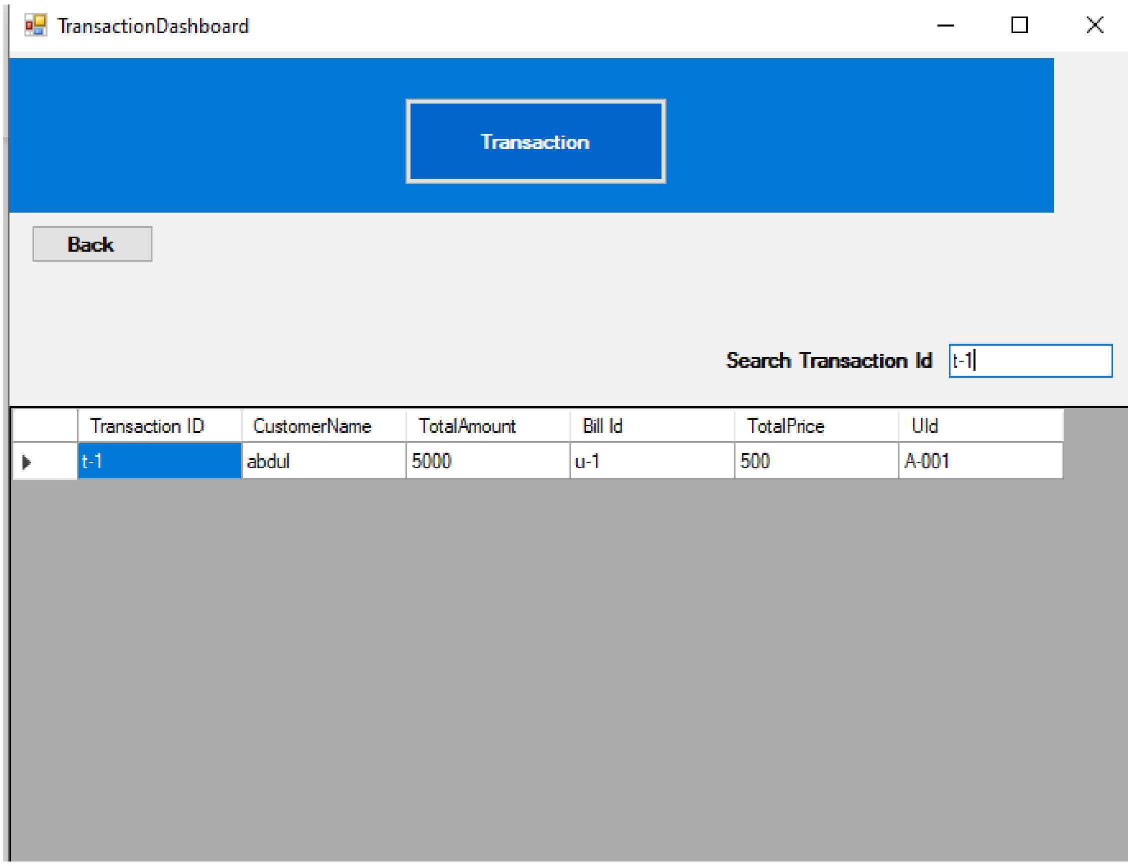

Exchange items
Available amount for SAMSUNG EVO 225GB SSD: 11
Enter Amount:
Enter

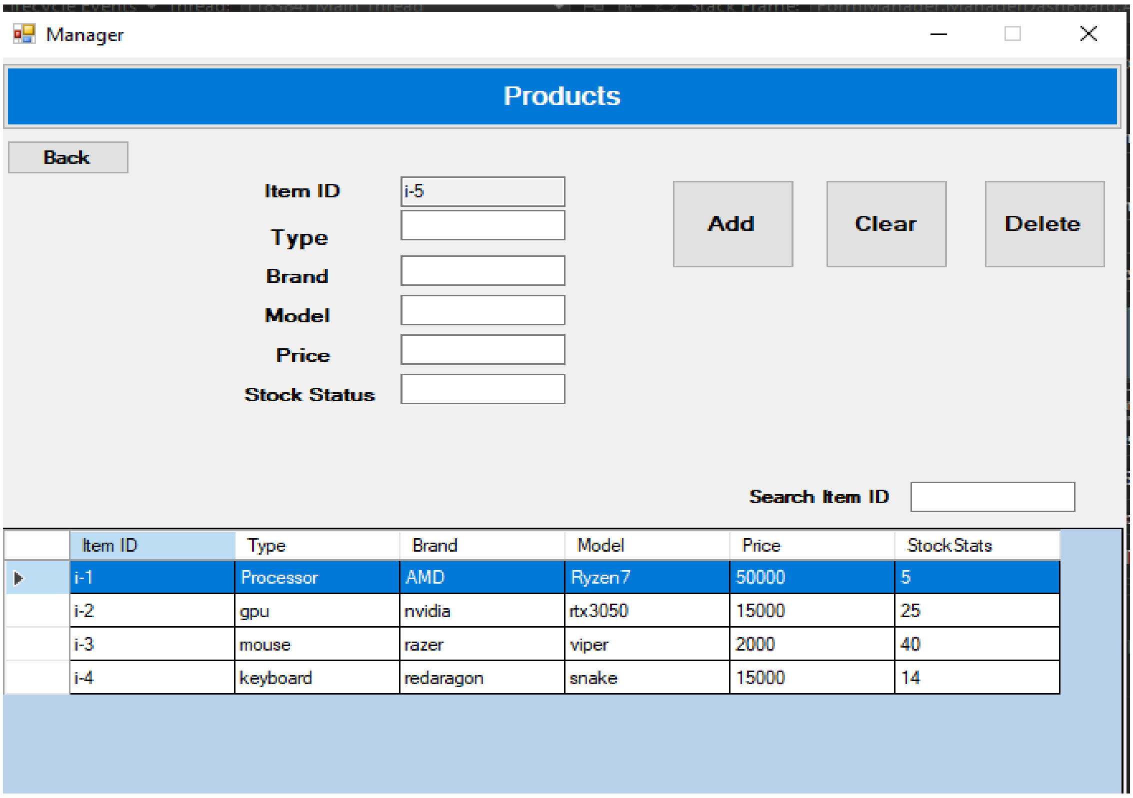

Update password
Update password
Currently logged in as:
S-001
Enter current password:
****
S/H
New password:
12345
S/H
Confirm password:
*****
S/H
Confirm

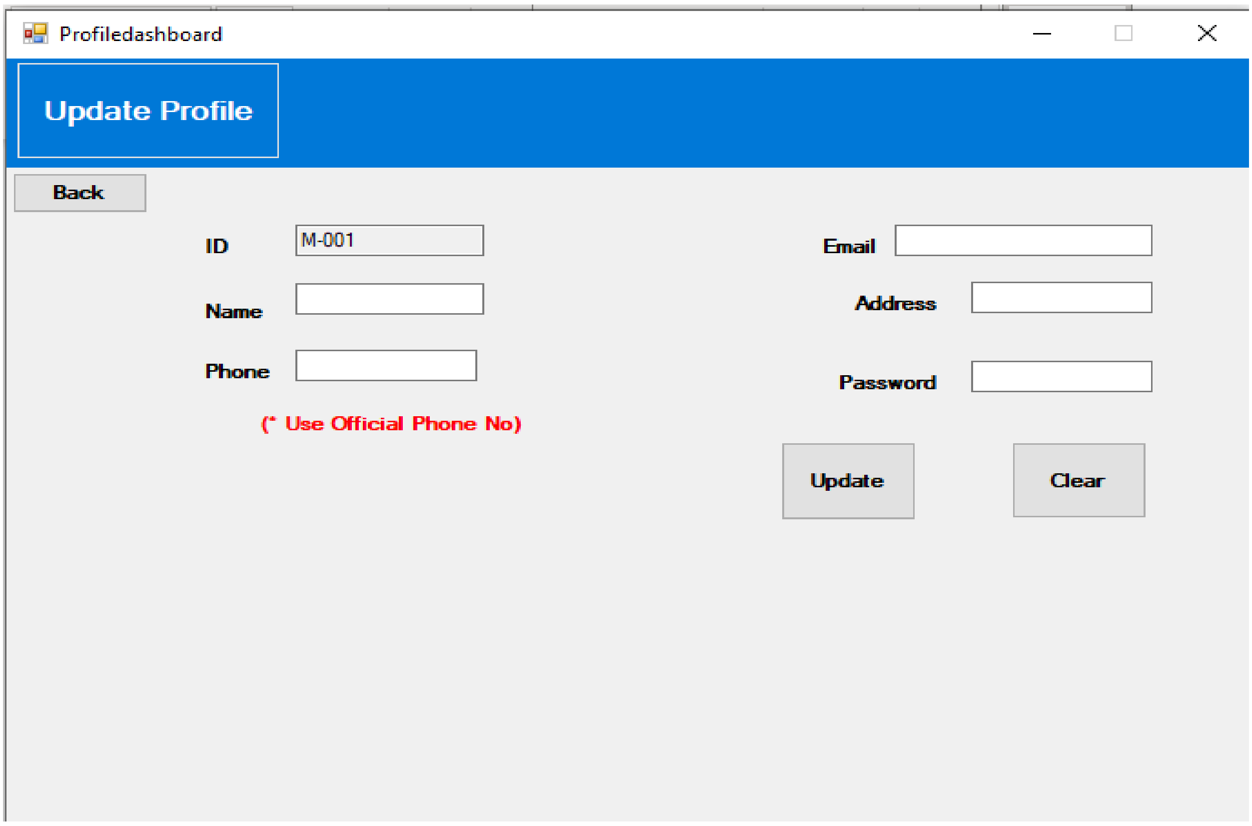

View receipt
Receipt
Serial ... Manuf... Model Item ty... Am... Total p...
1 AMD RYZE... Proces... 1 22000
2 Samsu... EVO 2... SSD 1 2000
3 GIGA... REVO ... RAM 1 1500
4 Intel i5-5230 Proces... 1 15000
Total items: 4
Total price with 10% discount (BDT): 36450
Customer name:
Tashdid
Customer contact:
1234
Purchase confirmed by: Salesman (Id: S-001)
Time of purchase: 12-19-2023
Okay

ID: 21-45833-3:


MainDashBoard
Welcome Manager
Items
Transaction
Update Profile
Log Out


TransactionDashboard
Transaction
Back
Search Transaction Id
t-1
Transaction ID
CustomerName
TotalAmount
Bill Id
TotalPrice
UId
t-1
abdul
5000
u-1
500
A-001


Manager
Products
Back
Item ID
i-5
Type
Brand
Model
Price
Stock Status
Add
Clear
Delete
Search Item ID
Item ID Type Brand Model Price Stock Stats
i-1 Processor AMD Ryzen7 50000 5
i-2 gpu nvidia rtx3050 15000 25
i-3 mouse razer viper 2000 40
i-4 keyboard redaragon snake 15000 14


Profiledashboard
Update Profile
Back
ID
M-001
Name
Phone
(* Use Official Phone No)
Email
Address
Password
Update
Clear

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

Project Requirements

Course Code and Title

CSC 2210 Object Oriented Programming 2

Credit

3 Credits (2hrs theory and 2hrs 20minutes Lab per week)

Nature

Core Course for CSE and COE

Prerequisite

CSC 2209: Object Oriented Programming 1

Semester

Spring 2025-2026

Project Requirements

1. <u>Project Title</u>

(Proposed project title)

2. <u>Project Description</u>

Example Structure:

- The system will support multiple types of users (e.g., Admin, Customer, Staff).
- Each user type will have specific functionalities.

User Roles and Features

User Type 1 (e.g., Admin)

- Feature 1
- Feature 2
- Feature 3
- Additional features as required

User Type 2 (e.g., Customer/User)

- Feature 1
- Feature 2
- Feature 3
- Additional features as required

(Add more user roles if needed)

3. <u>General Requirements</u>

3.1 **System Design & Architecture**

- The application must follow Object-Oriented Programming (OOP) principles:
    - Encapsulation
    - Inheritance
    - Polymorphism
    - Abstraction
- The system must be a desktop-based application (C#).
- Proper class design must be implemented (minimum 2 types of user classes).

3.2 **Database Requirements**

- Must use a relational database.
- Database should be normalized (at least 3NF).
- Implement CRUD operations (Create, Read, Update, Delete).
- Include a database connection class.

3.3 **Functional Requirements**

- Provide a search option accessible to all users.
- Implement login functionality:
    - Only ID and Password are required.
- All forms must be properly connected.
- Include a billing/record system (if applicable to the project).

3.4 **User Interface Requirements**

- Must include multiple forms.
- Apply proper form design principles:
    - Clear layout

    - User-friendly navigation
- Ensure desktop-based UI design consistency.

3.5 **Validation & Error Handling**

- Implement proper form validation (e.g., required fields, input format).
- Include exception handling throughout the application.

3.6 **Programming Practices**

- Proper use of:
    - Properties
    - Access modifiers (public, private, protected)
- Code should be modular and maintainable.

3.7 **Application Type**

- Must be a management-type application, such as:
    - Restaurant Management System
    - Library Management System
    - Student Management System
    - Inventory System

      (or similar)

4. Additional Notes

- The project should demonstrate clear use of OOP concepts in C#.
- Code readability, structure, and organization will be considered in evaluation.
- Extra features beyond requirements may improve grading if implemented properly.